DATA DICTIONARY

Laboratory Inventory Management Database

This data dictionary is based strictly on the provided Entity and Attribute Design, Primary Keys, and Foreign Keys. Data types below follow the SQL schema prepared for MySQL 8.x.

User

Attribute	Data Type	Key / Constraint	Description
user_id	INT	PK, AUTO_INCREMENT	Unique user identifier
role_id	INT	FK	Assigned role
username	VARCHAR(100)	UNIQUE, NOT NULL	Login identifier
password_hash	VARCHAR(255)	NOT NULL	Secure password hash
first_name	VARCHAR(100)	NOT NULL	First name
last_name	VARCHAR(100)	NOT NULL	Last name
email	VARCHAR(255)		Email, if used
status	VARCHAR(50)		Account status
created_at	DATETIME	NOT NULL	Creation date/time
updated_at	DATETIME	NOT NULL	Last update date/time

Role

Attribute	Data Type	Key / Constraint	Description
role_id	INT	PK, AUTO_INCREMENT	Unique role identifier
role_name	VARCHAR(100)	UNIQUE, NOT NULL	Role name
description	TEXT		Role description

Laboratory

Attribute	Data Type	Key / Constraint	Description
laboratory_id	INT	PK, AUTO_INCREMENT	Unique laboratory identifier
name	VARCHAR(150)	NOT NULL	Laboratory name
type	VARCHAR(100)		Laboratory type
location	VARCHAR(255)		Physical location
description	TEXT		Description
status	VARCHAR(50)		Laboratory status
created_at	DATETIME	NOT NULL	Creation date/time

User_Laboratory

Attribute	Data Type	Key / Constraint	Description
user_id	INT	PK, FK	References User
laboratory_id	INT	PK, FK	References Laboratory

Category

Attribute	Data Type	Key / Constraint	Description
category_id	INT	PK, AUTO_INCREMENT	Unique category identifier
name	VARCHAR(150)	UNIQUE, NOT NULL	Category name
description	TEXT		Description
status	VARCHAR(50)		Active/inactive

Inventory_Item

Attribute	Data Type	Key / Constraint	Description
item_id	INT	PK, AUTO_INCREMENT	Unique inventory identifier
item_name	VARCHAR(200)	NOT NULL	Item name
description	TEXT		Description
category_id	INT	FK, NOT NULL	Associated category
laboratory_id	INT	FK, NOT NULL	Assigned laboratory
item_type	VARCHAR(50)		Consumable or equipment
unit_of_measurement	VARCHAR(50)		Unit
quantity	DECIMAL(12,2)		Current quantity
minimum_stock_level	DECIMAL(12,2)		Low-stock threshold
condition	VARCHAR(100)		Current condition
location	VARCHAR(255)		Storage/location

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acquisition_date	DATE		Date acquired
acquisition_source	VARCHAR(255)		Source
remarks	TEXT		Additional information
status	VARCHAR(50)		Inventory status
created_at	DATETIME	NOT NULL	Date added
updated_at	DATETIME	NOT NULL	Last update
archived_at	DATETIME		Archive date, if applicable

Asset

Attribute	Data Type	Key / Constraint	Description
asset_id	INT	PK, AUTO_INCREMENT	Unique asset identifier
item_id	INT	FK, NOT NULL	Related inventory item/type
laboratory_id	INT	FK, NOT NULL	Current laboratory
status	VARCHAR(50)		Operational status
condition	VARCHAR(100)		Physical condition
remarks	TEXT		Additional information
created_at	DATETIME	NOT NULL	Registration date
updated_at	DATETIME	NOT NULL	Last update

Inventory_Transaction

Attribute	Data Type	Key / Constraint	Description
transaction_id	INT	PK, AUTO_INCREMENT	Unique transaction
item_id	INT	FK, NOT NULL	Affected item
user_id	INT	FK, NOT NULL	User performing transaction
transaction_type	VARCHAR(50)		Stock-in, stock-out, adjustment, or transfer
quantity	DECIMAL(12,2)		Quantity involved
transaction_date	DATETIME	NOT NULL	Date/time
reason	TEXT		Reason
source	VARCHAR(255)		Incoming source, if applicable
remarks	TEXT		Additional information

Borrowing_Request

Attribute	Data Type	Key / Constraint	Description
request_id	INT	PK, AUTO_INCREMENT	Unique request
requester_id	INT	FK, NOT NULL	Submitting user
laboratory_id	INT	FK, NOT NULL	Laboratory involved
purpose	TEXT		Purpose
request_date	DATETIME	NOT NULL	Request date
expected_return_date	DATETIME		Expected return
status	VARCHAR(50)		Workflow status
approved_by	INT	FK	Approver
approval_date	DATETIME		Approval/rejection date
approval_reason	TEXT		Approval/rejection reason
remarks	TEXT		Additional information

Borrowing_Request_Item

Attribute	Data Type	Key / Constraint	Description
request_item_id	INT	PK, AUTO_INCREMENT	Unique request detail
request_id	INT	FK, NOT NULL	Related request
item_id	INT	FK, NOT NULL	Requested item
requested_quantity	DECIMAL(12,2)		Requested quantity
remarks	TEXT		Additional information

Release

Attribute	Data Type	Key / Constraint	Description
release_id	INT	PK, AUTO_INCREMENT	Unique release
request_id	INT	FK, NOT NULL	Related request
released_by	INT	FK, NOT NULL	Releasing user
release_date	DATETIME	NOT NULL	Release date/time
remarks	TEXT		Additional information

Release_Item

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Attribute	Data Type	Key / Constraint	Description
release_item_id	INT	PK, AUTO_INCREMENT	Unique release detail
release_id	INT	FK, NOT NULL	Related release
item_id	INT	FK, NOT NULL	Released item
asset_id	INT	FK, NULL	Individual asset, when applicable
quantity_released	DECIMAL(12,2)		Quantity released
remarks	TEXT		Additional information

Return

Attribute	Data Type	Key / Constraint	Description
return_id	INT	PK, AUTO_INCREMENT	Unique return
request_id	INT	FK, NOT NULL	Related request
processed_by	INT	FK, NOT NULL	Processing user
return_date	DATETIME	NOT NULL	Return date
remarks	TEXT		Additional information

Return_Item

Attribute	Data Type	Key / Constraint	Description
return_item_id	INT	PK, AUTO_INCREMENT	Unique return detail
return_id	INT	FK, NOT NULL	Related return
item_id	INT	FK, NOT NULL	Returned item
asset_id	INT	FK, NULL	Individual asset, when applicable
quantity_returned	DECIMAL(12,2)		Quantity returned
missing_quantity	DECIMAL(12,2)		Quantity missing
damaged_quantity	DECIMAL(12,2)		Quantity damaged
condition	VARCHAR(100)		Condition upon return
remarks	TEXT		Additional information

Damage_Loss_Record

Attribute	Data Type	Key / Constraint	Description
damage_loss_id	INT	PK, AUTO_INCREMENT	Unique damage/loss record
item_id	INT	FK, NULL	Affected inventory item
asset_id	INT	FK, NULL	Affected asset
record_type	VARCHAR(50)		Damage or loss
date_recorded	DATETIME	NOT NULL	Date recorded
description	TEXT		Description
reported_by	INT	FK, NOT NULL	Reporting user
current_condition	VARCHAR(100)		Current condition
action_taken	TEXT		Action taken
remarks	TEXT		Additional information

Maintenance_Record

Attribute	Data Type	Key / Constraint	Description
maintenance_id	INT	PK, AUTO_INCREMENT	Unique maintenance record
asset_id	INT	FK, NOT NULL	Maintained asset
reported_issue	TEXT		Reported issue
date_reported	DATETIME	NOT NULL	Report date
maintenance_action	TEXT		Work performed
responsible_person	VARCHAR(200)		Responsible person
date_completed	DATETIME		Completion date
maintenance_status	VARCHAR(50)		Status
remarks	TEXT		Additional information

Audit_Log

Attribute	Data Type	Key / Constraint	Description
audit_id	INT	PK, AUTO_INCREMENT	Unique audit record
user_id	INT	FK, NOT NULL	User performing action
timestamp	DATETIME	NOT NULL	Date/time
action	VARCHAR(100)		Action performed
affected_entity	VARCHAR(100)		Affected entity
record_id	INT		Affected record ID
previous_value	TEXT		Previous value, where applicable
new_value	TEXT		New value, where applicable

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reason	TEXT		Reason, where applicable

Note: The source specification defines the attributes, keys, purposes, and relationships. It does not define exact SQL data types, lengths, or default values; therefore, the data types shown here correspond to the MySQL 8.x implementation prepared from that specification.

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